

## BINLEVEL · DESIGN DECISION RECORD

# BinLevel recommendations accepted

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## DOCUMENT CONTROL

|             |                                                               |         |                |
|-------------|---------------------------------------------------------------|---------|----------------|
| Document ID | BNL-DDR-002                                                   | Version | 0.1            |
| Status      | <b>Draft</b>                                                  | Date    | 2026-09-25     |
| Author      | Amish Chadha                                                  | License | CERN-OHL-S-2.0 |
| Project TRL | <b>TRL 3</b> Proof of concept                                 |         |                |
| Repository  | github.com/BoujeeEnjinia1701/binlevel                         |         |                |
| Source      | docs/decisions/0002-recommendations-accepted.md @ uncommitted |         |                |

## REVISION HISTORY

| Version | Date       | Author       | Change                                                                            |
|---------|------------|--------------|-----------------------------------------------------------------------------------|
| 0.1     | 2026-09-25 | Amish Chadha | Record Amish's acceptance of all recommendations and the changes made in the repo |

# 0002: Recommendations accepted

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- **Date:** 2026-09-25
- **Status:** accepted. Every item below that carried a recommendation is "Decided by Amish, 2026-09-25: go with recommendation". Items without a recommendation remain "Proposed, awaiting Amish".

## ■ Context

On 2026-09-25 Amish wrote, in chat: "i accept all your recommendations, go with them across all repos." BinLevel had ten items adopted for TRL 3 under his earlier instruction and open for his review (BNL-DDR-001, D1 to D10), four new items raised by the sizing note BNL-CAL-001 v0.1 with a recommendation each, one suggestion on the reporting interval, and two items with no recommendation (O1, O2). Where a recommendation offered several options, the recommended option is the decision. TRL 4 remains on hold by Amish's instruction, and the repo stays at TRL 3.

## ■ Decision

*Table 1. Items decided by Amish, 2026-09-25: go with recommendation.*

| #         | Item                                                                                                                                                                                                                               | Decision                                                                                               | What changed in the repo                                                                                                                                                                                                                                                                                                 |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| D1 to D10 | TRL 2 review items (BNL-DDR-001): communal containers first, ultrasonic plus ToF, C cell, LoRaWAN on RAK3172, FieldNode and TwinKit reuse, 80 % and 1 h defaults, heat alert, lid mounting, stock IP67 enclosure, no budget change | As recommended                                                                                         | Status wording in BNL-DDR-001 (v0.2), BNL-PRB-001, BNL-PRC-001 and README; no design change                                                                                                                                                                                                                              |
| N1        | R16 mass: (a) 2 mm aluminium bracket, (b) relax R16 to 450 g, (c) perforated stainless                                                                                                                                             | (a) 2 mm 5052-class aluminium bracket                                                                  | cad/src/model.py plate 1.5 mm stainless to 2.0 mm aluminium; STEP and STL re-exported; BOM item 2 renamed and respecified (price unchanged at \$5.00); drawing BNL-DWG-001 to Rev P2; BNL-CAL-001 v0.2: unit mass 435 g to 334 g, envelope below the lid 61.0 to 61.5 mm, shock load 85 to 66 N; R16 from not met to met |
| N2        | R6 upper limit                                                                                                                                                                                                                     | Raise from 60 to 70 °C; name the RAK3172-T (-40 to 85 °C) for cold sites                               | BNL-REQ-001 R6 target; BNL-CAL-001 speed-of-sound range now -20 to 70 °C (uncompensated error up to +8.2 %, was +6.6 %); R6 stays at risk (3 K margin, transducer rating unconfirmed)                                                                                                                                    |
| N3        | R9 rate-of-rise rule                                                                                                                                                                                                               | Count the 15 K rise in 15 min only when the unit is already above 50 °C; keep the 70 °C absolute alert | BNL-REQ-001 R9 target; BNL-PRC-001 firmware rule; BNL-CAL-001 new checks H3 and H4 (unit about 43 °C under cloud, below the gate; worst rise above the gate 10.9 K); R9 from at risk to met on paper                                                                                                                     |
| N4        | 11-byte payload and 5 min temperature reads                                                                                                                                                                                        | Keep both, as applied at TRL 3                                                                         | Status wording only                                                                                                                                                                                                                                                                                                      |
| N5        | Reporting interval stretch (suggestion in BNL-CAL-001 v0.1)                                                                                                                                                                        | Stretch the routine interval to 2 h at SF12 only; SF11 stays hourly (21.4 s a day, inside TTN's 30 s)  | BNL-REQ-001 R3, BNL-PRC-001, BNL-CAL-001 section E, concept media key figures; worst-case battery life unchanged at 16.3 years                                                                                                                                                                                           |
| N6        | Budget                                                                                                                                                                                                                             | No change was recommended                                                                              | budget_usd stays \$60; parts total stays \$55.00                                                                                                                                                                                                                                                                         |

Table 2. Items still open.

| #  | Item                                                                                                                                                                         | Status                   |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| O1 | Steel containers: (a) external lid antenna variant (about \$8, total \$63.00, over budget) or (b) exclude steel containers from the first pilot. No recommendation was made. | Proposed, awaiting Amish |
| O2 | First partner and region for co-design and a pilot; also sets the radio plan (EU868, US915 or IN865). No preference stated.                                                  | Proposed, awaiting Amish |

Table 3. Decided but on hold or outside this repo.

| Item                                                                                                                                                                                                           | Status                                                                                        |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| TRL 4 work named in the review note (bench build, lab test report, ranging on real waste, current and cold-start measurements, link loss in HDPE and steel containers, lid temperature in sun, shake and drop) | Decided as the next step, on hold: TRL 4 is on hold by Amish's instruction                    |
| FieldNode's 20-byte payload exceeds the 11-byte US915 DR0 limit found here, and FieldNode stretches its interval from SF10                                                                                     | Cross-repo action for FieldNode, listed in <code>docs/REVIEW.md</code> ; FieldNode not edited |

## ■ Consequences

- Requirement status (BNL-CAL-001 v0.2): not met 2 (R5 IP69K; R10 in steel containers), at risk 2 (R2, R6), not verifiable at TRL 3 2 (R7, R12), met 10 (R1, R3, R4, R9, R14, R16 by calculation; R8, R11, R13, R15 by design). Before this record: not met 3, at risk 3, met 8.
- Controlled documents bumped: BNL-PRB-001 v0.4, BNL-PRC-001 v0.4, BNL-REQ-001 v0.4, BNL-CAL-001 v0.2, BNL-DDR-001 v0.2; drawing BNL-DWG-001 Rev P2.
- The stainless bolts now pass through an aluminium plate. Galvanic isolation (anodized plate or insulating washers) is a review suggestion, not a decision.
- `trl: 3` and `trl_target: 3` are unchanged.